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PROJECT: CLOSED-LOOP pH REGULATION SYSTEM FOR HYDROPONIC APPLICATIONS

ECE 0323.1 - FEEDBACK AND CONTROL SYSTEM (LABORATORY)

BS ECE 3-1

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Abstract

This project presents the design, implementation, and testing of a closed-loop pH regulation system intended for hydroponic water management. The system maintains a user-defined pH setpoint within a programmed deadband by sensing the water's pH with a glass-electrode probe and dispensing controlled volumes of acidic and basic solutions through two peristaltic dosing pumps. The controller is an Arduino Uno running an on-off control algorithm with deadband and dead-time compensation, and a 16-by-2 liquid crystal display provides real-time feedback of the measured pH, the setpoint, and the controller state. The pH sensor is calibrated through a two-point linear procedure using standard buffer solutions at pH 4.01 and pH 6.86, with verification across pH 9.18 and distilled water. The completed system holds the reservoir pH steady within the deadband when undisturbed and responds to manually introduced acid and base disturbances by issuing short corrective doses followed by mixing-wait periods, returning the pH toward the setpoint. The work demonstrates that low-cost, locally available components can implement the essential feedback control loop taught in classical control theory and provides a foundation for future enhancements including active mixing, temperature compensation, and proportional or integral control.

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1. Introduction

1.1 Background

Hydroponics is a soilless cultivation method in which plants draw nutrients directly from an aqueous solution circulated around the roots. Because the plant has no soil to buffer chemical fluctuations, the chemistry of the nutrient water becomes the single most important variable in the system. Of the parameters that govern plant health in such systems, the pH of the water is among the most critical.

The pH of a nutrient solution determines whether dissolved nutrients are in chemical forms that plants can absorb. When the pH drifts outside the optimal range, certain nutrients become unavailable to the plant even though they remain physically present in the water. For most leafy hydroponic crops such as lettuce, the literature recommends maintaining a pH of approximately 5.5 to 6.5. Outside of this band, growth slows, deficiency symptoms appear, and yields fall.

Small backyard and educational hydroponic setups commonly rely on manual pH checks using handheld meters or test strips, followed by hand addition of pH-up or pH-down adjusting solutions. This approach is labor-intensive, prone to inconsistency, and not feasible for unattended operation. Commercial automated controllers exist but tend to be priced for industrial users and are not readily accessible to students or small growers. There is therefore a clear case for a low-cost, locally constructible automated pH regulation system that demonstrates the principles of feedback control in an applied context.

1.2 Problem Statement

The challenge addressed by this project is to design and implement an automated system that maintains the pH of a water reservoir within a user-defined band using closed-loop feedback control. The system must sense the pH continuously, compare the measurement against a setpoint, and dispense acidic or basic correction solution in response, all without operator intervention beyond initial setup and the replenishment of dosing reagents.

1.3 Objectives

The general objective of this project is to design, build, and demonstrate a closed-loop pH regulation system suitable for hydroponic applications, using off-the-shelf microcontroller hardware and standard chemical reagents.

Specific objectives are as follows.

- To implement analog pH sensing using a commercial glass-electrode probe with signal conditioning, including the formulation and execution of a proper two-point calibration procedure with traceable buffer solutions.
- To design and assemble an actuation subsystem capable of delivering measured volumes of acidic and basic dosing fluids through peristaltic pumps under microcontroller control.
- To implement an on-off control algorithm with a programmed deadband and a mandatory mixing-and-settling wait period after each dose, addressing the nonlinearity and dead time inherent to pH systems.
- To provide a real-time human interface through a liquid crystal display so that the current pH, the setpoint, and the controller state can be observed at any time during operation.

- To assemble the system from low-cost components available in the local market and to verify its operation through controlled disturbance tests.

1.4 Scope and Limitations

This project is a proof-of-concept implementation. The system is demonstrated on a small reservoir of distilled water rather than on a fully operational hydroponic setup with living plants and circulating nutrient solution. Only pH is regulated; the system does not measure or control electroconductivity, nutrient concentration, water temperature, dissolved oxygen, or water level. For demonstration purposes, household white vinegar serves as the acidic dosing agent and a sodium bicarbonate solution prepared from food-grade baking soda serves as the basic dosing agent, in place of commercial pH-up and pH-down products. Active mixing is not implemented; light manual agitation is used between observations. Temperature compensation is not implemented because all testing is conducted at approximately laboratory room temperature. The controller is an on-off scheme with deadband; proportional, integral, and derivative control are identified as future work.

2. Theoretical Background

2.1 pH and the Glass Electrode

pH is a logarithmic measure of the hydrogen ion activity in an aqueous solution, defined on a scale from 0 to 14 where 7 is neutral. Values below 7 indicate acidity and values above 7 indicate alkalinity. Because the scale is logarithmic, a change of one pH unit corresponds to a tenfold change in hydrogen ion concentration; this logarithmic relationship is a fundamental reason that pH systems exhibit strong nonlinearity and resist simple linear control.

The standard instrument for measuring pH is the glass-electrode probe, also known as a combination pH electrode. It consists of two half-cells. The sensing half-cell is a thin glass membrane that develops a small potential difference across itself in proportion to the hydrogen ion activity of the solution it is immersed in. The reference half-cell, typically silver and silver chloride immersed in a 3-molar potassium chloride solution, provides a stable reference potential. The output of the probe is the difference between these two potentials. According to the Nernst equation, this output voltage varies linearly with pH at a theoretical slope of 59.16 millivolts per pH unit at a temperature of 25 degrees Celsius.

2.2 Sensor Calibration

Real glass electrodes deviate from the Nernst ideal because of manufacturing tolerances in the membrane, aging of the reference electrode, and contamination of the reference junction. The data sheet of the probe used in this project specifies a zero-point tolerance of plus or minus 0.5 pH units, meaning that the raw probe voltage may correspond to a pH up to half a unit away from the actual pH of the solution. Calibration is therefore mandatory.

A two-point linear calibration was used. With the probe placed in a buffer solution of known pH, the output voltage is recorded once the reading has stabilized. A second known buffer at a different pH yields a second voltage reading. The two ordered pairs of voltage and pH are sufficient to determine the slope and intercept of a linear equation that maps any subsequent voltage measurement to a pH value. In this project, buffers at pH 4.01 and pH 6.86 were used as the primary calibration points, with additional verification against pH 9.18 buffer and distilled water.

2.3 Feedback Control

A feedback control system continuously measures the state of a process variable, compares it against a desired value called the setpoint, and adjusts a manipulated variable through an actuator in order to drive the process variable toward the setpoint. The defining property of feedback control, as distinguished from open-loop control, is that the sensor measurement is fed back into the control decision; the actuator's action depends on the present state of the plant rather than on a fixed, predetermined schedule.

In this project, the controlled variable is the pH of the reservoir water, the setpoint is the target pH defined in the controller firmware, the manipulated variable is the volume of acidic or basic dosing solution introduced into the reservoir, the actuators are the two peristaltic pumps, the plant is the reservoir of water itself, and the sensor is the glass-electrode pH probe submerged in the reservoir. Together, these elements form a closed loop.

2.4 Selection of Control Strategy

Several control strategies were considered for this project. A proportional controller adjusts the actuator output in linear proportion to the magnitude of the error between the setpoint and the measurement. While conceptually simple, a pure proportional controller leaves a steady-state offset and is sensitive to the strong nonlinearity of pH systems. A proportional-integral or proportional-integral-derivative controller provides improved transient and steady-state performance but requires careful tuning of multiple gain parameters; tuning is particularly challenging for pH systems because the plant's behavior near neutral pH is highly nonlinear.

On-off control, sometimes called bang-bang control, switches the actuator fully on or fully off depending on whether the measurement lies above or below the setpoint. In its pure form, on-off control causes continuous chattering of the actuator and oscillation of the controlled variable around the setpoint. The chattering can be eliminated by introducing a deadband: a band of values around the setpoint inside which no control action is taken. Outside the deadband, the controller activates the appropriate actuator until the controlled variable returns to within the band.

For this project, on-off control with a deadband was selected for three reasons. First, it does not require tuning of continuous-gain parameters, which is well suited to the project's development timeline. Second, it accommodates the nonlinearity of pH without requiring an explicit plant model. Third, the chosen actuators are peristaltic pumps that physically operate in an on-off manner, making the algorithmic on-off control a natural match for the hardware. The control strategy also incorporates a mandatory dead-time wait after each dose, during which no further control action is taken, to allow the dose to mix into the reservoir and for the sensor to register the new equilibrium pH.

2.5 Dead Time and the Need for a Mixing Wait

When a dose of acid or base is added to a reservoir, several seconds pass before the dose has dispersed throughout the volume and before the probe registers the resulting change in pH. If the controller were permitted to react again before this dispersion was complete, it would over-correct based on stale information, producing overshoot and oscillation. The mandatory wait period after each dose is the simplest form of dead-time compensation: it forces the controller to pause until the previous correction has taken effect before evaluating the next decision. In a more advanced implementation, this delay could be replaced by an explicit Smith predictor or model-based dead-time compensator, but for the demonstration scope of the present project a fixed wait is sufficient.

3. System Design

3.1 System Overview

The system consists of six functional subsystems: a microcontroller running the control firmware, a pH sensor providing the feedback signal, two peristaltic pumps acting as dosing actuators, a relay module that switches the pumps on command from the microcontroller, a liquid crystal display that presents the system state to the operator, and a power conditioning subsystem that derives the necessary supply voltages from a single 12-volt source. The reservoir, a small plastic container holding the water to be regulated, sits at the center of the arrangement, with the pH probe immersed in it and the outlet tubes of the dosing pumps positioned to discharge into it.

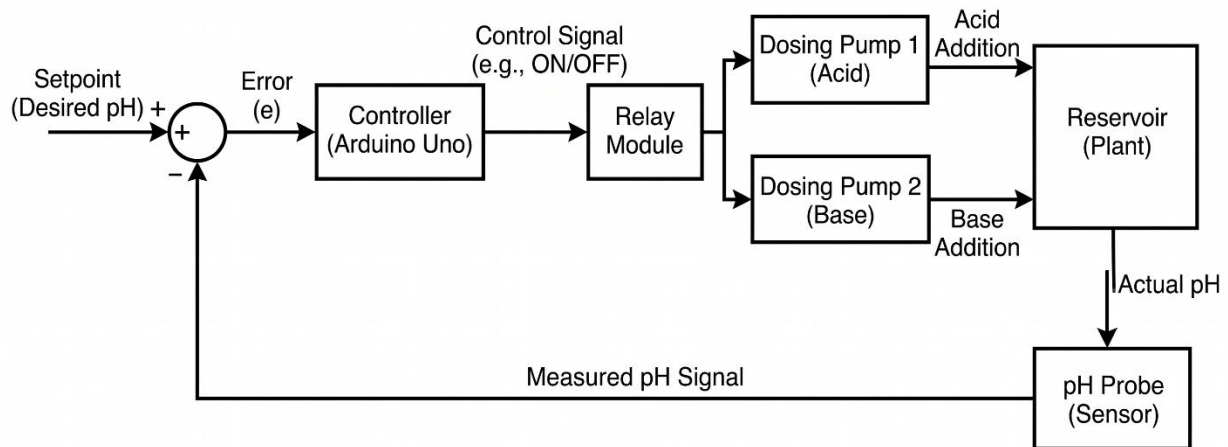


Figure 1. Closed-loop block diagram of the pH regulation system

3.2 Hardware Components

The hardware components were selected on the basis of cost, local availability, and suitability for the proof-of-concept scope. Table 1 lists each component with its specifications.

Component	Specifications	Quantity
Arduino Uno	ATmega328P microcontroller, 16 MHz clock, 10-bit ADC, powered through barrel jack at 12 volts	1
DFRobot SEN0161 Gravity Analog pH Sensor Kit	Glass-electrode probe (E-201 series) with BNC connector and signal conditioning board; 0-14 pH measurement range; 0-3 volt analog output	1
INTLLAB 12V DC peristaltic pump	5 watt power rating; food-grade silicone tubing; switched on and off through the relay module	2
2-channel relay module	5-volt logic side, screw-terminal switching contacts, toggle-pulse operation	1
16x2 I2C LCD module	HD44780-compatible character display with PCF8574 I2C backpack at address 0x27	1

LM2596 buck converter	Adjustable step-down DC-DC converter, input 12 volts, output adjusted to 5 volts to power the LCD, relay logic, and sensor module	1
12V 2A DC power adapter	Center-positive barrel plug, regulated linear or switching supply	1
pH buffer powder sachets	Premixed powder for pH 4.01, pH 6.86, and pH 9.18 calibration solutions; reconstituted in 250 mL of distilled water each	1 set
3M potassium chloride storage solution	Supplied with the probe for membrane hydration and long-term storage	1 bottle
Food-grade silicone tubing	3 mm inner diameter; used for the dosing fluid paths through the peristaltic pumps	Approx. 2 m
White vinegar (Datu Puti)	Acetic acid solution serving as the pH-down dosing reagent (approximate pH 2.4)	1 bottle
Sodium bicarbonate (baking soda)	Dissolved in distilled water at approximately 6 percent concentration by mass to serve as the pH-up dosing reagent (approximate pH 8.5 to 9.0)	1 small pack
Distilled water	Used for buffer mixing, probe rinsing, and the test reservoir	Approx. 1 L
Plastic reservoir container	Approximately 1 liter capacity, food-grade polypropylene	1
Breadboard and jumper wires	Solderless prototyping breadboard with assorted male-to-male and male-to-female jumpers	As needed

Table 1. Hardware bill of materials

3.3 Power Architecture

The entire system runs from a single 12-volt DC adapter. This single supply is distributed in three directions. First, the 12-volt rail is wired into the Arduino's barrel jack, where the Arduino's onboard linear regulator further steps it down to provide internal 5-volt logic power to the microcontroller. Second, the 12-volt rail feeds an LM2596 buck converter that produces a separate, regulated 5-volt output used to power the pH sensor module, the LCD module, and the logic side of the relay module. Third, the 12-volt rail is wired to the common terminals of the relay so that when a relay channel switches, it routes 12 volts to the corresponding pump.

The decision to use two independent 5-volt regulators is intentional. The Arduino is powered through its own onboard regulator, isolating the microcontroller from any electrical noise generated by the relay switching and the inrush current of the pumps, while the external buck converter handles the larger and noisier peripheral load. All grounds in the system are tied together at a common point, providing a clean reference for the analog input. Feeding the Arduino through

its barrel jack rather than directly through its 5-volt pin avoids bypassing the onboard regulator and removes the risk of overvolting the logic in the event that the buck converter output ever drifts.

3.4 Signal Interconnections

The microcontroller communicates with the rest of the system through five signal lines: one analog input from the pH sensor, two digital outputs to the relay control inputs, and two I2C lines to the LCD. Table 2 summarizes the pin assignment.

Arduino Pin	Function	Connection
A0	Analog input	pH sensor module analog output
A4 (SDA)	I2C data	LCD module SDA
A5 (SCL)	I2C clock	LCD module SCL
D7	Digital output	Relay channel 1 input, controls pH-down pump
D8	Digital output	Relay channel 2 input, controls pH-up pump
D13	Digital output	Onboard LED, used as a heartbeat indicator
GND	Common ground	Tied to all module grounds and 12-volt supply ground
Barrel jack	12-volt input	Direct from the 12-volt supply

Table 2. Arduino Uno pin assignments

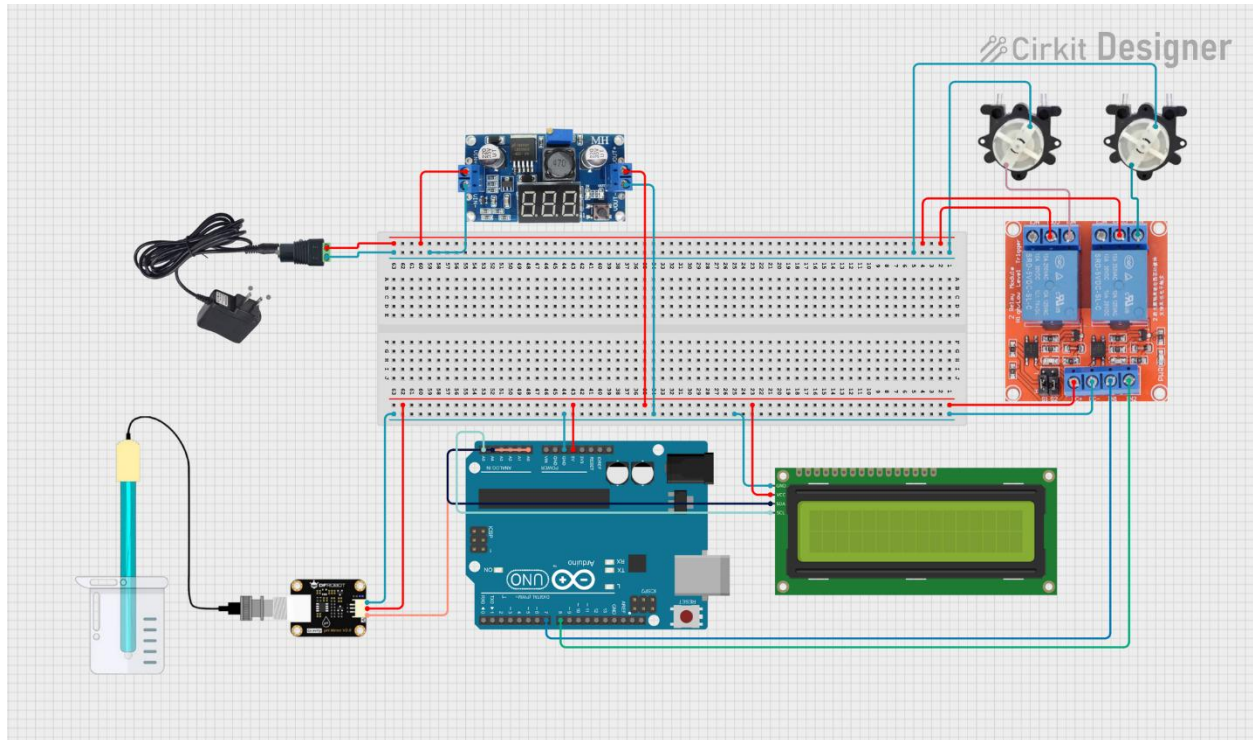


Figure 2. Wiring schematic of the complete hardware setup

3.5 Software Architecture

The control firmware runs as a single non-blocking loop. Each iteration of the loop performs five actions in succession. First, the analog input from the pH sensor is sampled at a fixed rate and stored in a circular buffer for averaging. Second, a trimmed-mean average of the buffer is computed and converted to a pH value using the calibration constants stored in the firmware. Third, the resulting pH value, the setpoint, and the current controller state are displayed on the LCD and also transmitted over the serial port for logging. Fourth, the controller state machine evaluates whether a dosing action is required based on the current pH and the deadband, and transitions through dosing and waiting states as needed. Fifth, an onboard LED is toggled once per second as a heartbeat indicator confirming that the loop is alive.

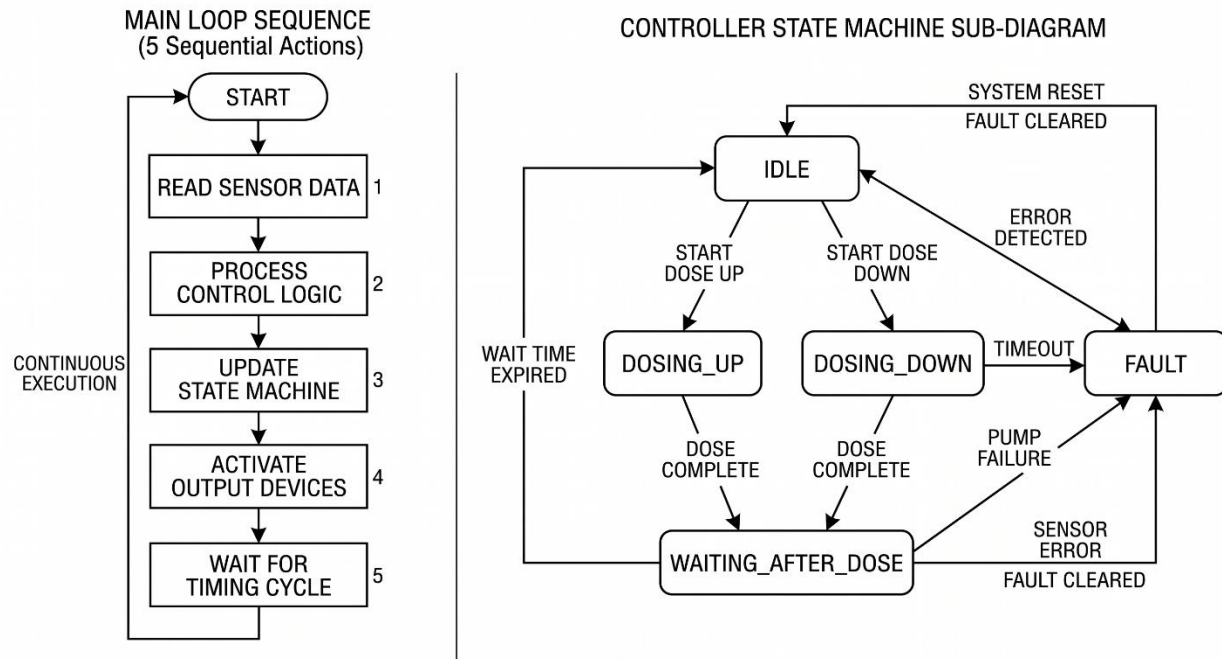


Figure 3. Software control flowchart

3.6 Safety and Fault Handling

Several safety provisions are incorporated into the firmware. On startup, the relay control pins are explicitly driven to their inactive logic level before the pinMode call is issued, eliminating the possibility of a transient pump activation during boot. During operation, every pH reading is checked against the physically meaningful range of zero to fourteen; if a reading falls outside this range, the system enters a FAULT state in which both pumps are commanded off and no control action is taken. The system automatically returns to normal operation when the reading recovers. A minimum dose interval, implemented as an independent check at the entry of the IDLE state, prevents successive doses from occurring closer in time than a configured floor, providing defense-in-depth alongside the primary dead-time mixing wait.

4. Implementation

4.1 Hardware Assembly

The assembly proceeded in stages, with each subsystem verified independently before integration. The Arduino, LCD, pH sensor module, and relay module were first arranged on a breadboard and connected to a temporary 5-volt source for basic sensor and display verification. After confirming that the LCD initialized correctly at I2C address 0x27 and that the pH sensor produced sensible voltage readings with the probe immersed in storage solution, the buck converter and the 12-volt supply were added, and the two peristaltic pumps were wired through the relay channels. Silicone tubing was routed from the pump inlets into the two dosing-fluid bottles and from the pump outlets to drape into the reservoir from above.

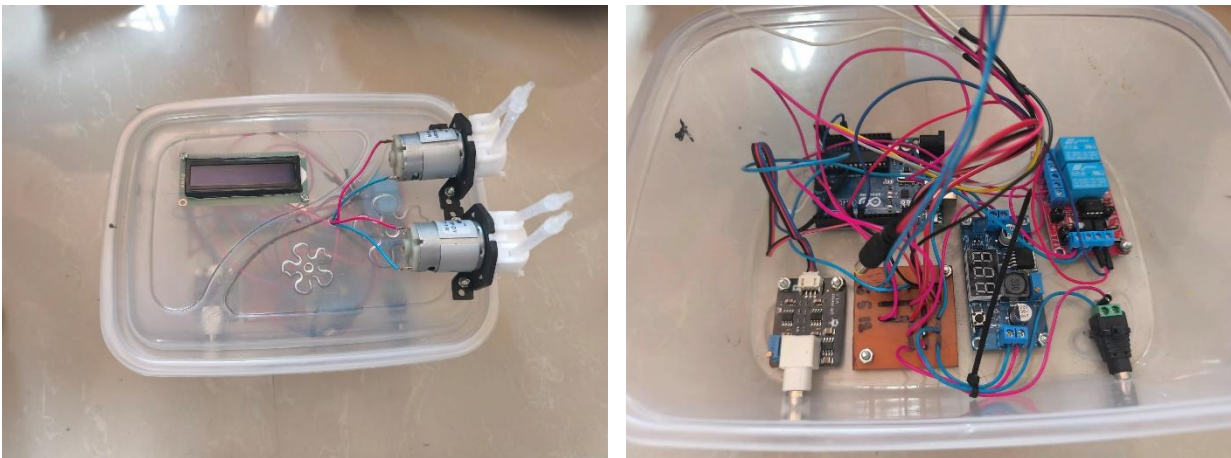


Figure 4. Photograph of the assembled prototype

4.2 Software Implementation

The control firmware is written in C++ using the Arduino IDE. Two external libraries are used: the Wire library for I2C communication with the LCD backpack, and the LiquidCrystal_I2C library for character output. The firmware is organized into clearly separated sections for configuration constants, helper functions, setup, and the main loop. Pump state is tracked in software because the relay module operates with toggle-pulse semantics and provides no state feedback. The full source listing is provided in Appendix B.

4.3 Relay Toggle-Pulse Handling

The relay module used in this project operates with toggle-pulse semantics rather than the level-driven semantics common to many relay modules. Each low-going pulse on a relay input pin inverts the state of the corresponding relay contact, regardless of the previous state. This required a small but important software adaptation: instead of simply asserting a logic level to turn a pump on or off, the firmware maintains a boolean variable for each pump's tracked state and pulses the relay only when the desired state differs from the current tracked state. At system boot the pumps are observed to be physically off, so the tracked state variables are initialized to false and no startup pulse is required.

4.4 Calibration Procedure

The pH sensor was calibrated through a two-point linear procedure using standard buffer solutions. Before calibration, the probe was hydrated by immersion in 3-molar potassium chloride storage solution for over eight hours to ensure the glass membrane was fully wetted and responsive. After initial testing revealed that the probe exhibited reduced sensitivity, an additional conditioning step was performed by soaking the probe in pH 4.01 buffer for approximately 45 minutes, which restored the probe's response across the measurement range.

For the calibration measurements proper, the probe was first rinsed with distilled water and blotted dry with a clean tissue, then submerged in the pH 4.01 buffer until the reading stabilized. The stable voltage and corresponding pH were recorded. The probe was then rinsed and blotted again and transferred to the pH 6.86 buffer for a second measurement. The recorded ordered pairs were used to derive the slope and offset of a linear calibration equation of the form $\text{pH} = \text{PH_SLOPE} \times \text{voltage} + \text{PH_OFFSET}$. The resulting calibration constants, namely a slope of 3.5 and an offset of 0.06, are stored in the controller firmware. The accuracy of the resulting calibration was verified by submerging the probe in pH 9.18 buffer and in distilled water and confirming that the reported pH agreed with the known values.

Test condition	Stability	Recorded voltage	Reported pH	Notes
pH 4.01 buffer (acid end)	Stable	1.20 V	4.27 (predicted by firmware)	Used as low calibration point; small acid-side bias is discussed in Section 7.
pH 6.86 buffer (near neutral)	Stable	Used in calibration computation	6.86 (target)	Primary near-neutral calibration point.
pH 9.18 buffer (verification)	Stable	2.61 V	9.20 (firmware output)	Verification at the basic end; agreement within 0.02 pH.

Table 3. Calibration measurement results

5. Testing Methodology

5.1 Test Setup

All testing was conducted on a benchtop in the laboratory at approximately room temperature. The test reservoir was a clean plastic container filled with approximately 500 milliliters of distilled water. The pH probe was submerged in the reservoir such that the glass bulb and the side reference junction were both fully immersed. The outlet tubes of the two pumps were positioned to drape into the reservoir from above, while their inlet tubes drew from separate small bottles labeled pH-DOWN, containing white vinegar at full strength, and pH-UP, containing a sodium bicarbonate solution prepared by dissolving approximately one tablespoon of food-grade baking soda in 250 milliliters of distilled water.

Because no active mixing device was used, the reservoir water was lightly agitated by hand for approximately five seconds before recording observations following any disturbance or dose. This manual agitation was sufficient to bring localized pockets of recently dosed solution into contact with the probe and to produce a representative reading.

5.2 Controller Configuration

The controller configuration parameters used during testing are summarized in Table 4. These values were arrived at through tuning trials. The dose duration was reduced from initial trial values to 10 milliseconds because peristaltic pump pulses of this brief duration were observed to deliver a droplet sufficient to shift the reservoir pH by approximately 0.1 to 0.2 units, which provided a visible response without overshooting the deadband. The mixing-wait was reduced from longer trial values to 10 seconds, which was found to be sufficient given the small reservoir volume and the high effective concentration of the dosing reagents.

Parameter	Value	Rationale
Setpoint	pH 6.0	Target pH; chosen to lie within the recommended hydroponic range.
Deadband (total width)	0.4 pH	Dosing is triggered when measurement falls outside [5.8, 6.2].
Dose duration	10 ms	Single-pulse pump activation; delivers a droplet shifting pH by approximately 0.1-0.2.
Dead-time mixing wait	10 s	Minimum time after a dose before any further evaluation.
Minimum dose interval	10 s	Independent safety check; mirrors the mixing wait.
pH sanity check range	0 to 14	Readings outside this range trigger FAULT state and disable pumps.

Table 4. Controller configuration parameters

5.3 Test Scenarios

Three test scenarios were used to characterize the system's behavior. The first scenario, steady-state stability, placed the reservoir at approximately the setpoint with no external disturbance, allowing observation of natural drift and noise in the reading. The second scenario, single-droplet response, manually agitated the reservoir over a five-second interval and observed the resulting small pH change produced by a single 10-millisecond dose. The third scenario, large disturbance response, deliberately perturbed the reservoir by pouring an unmeasured but visibly significant amount of vinegar to drive the pH downward, and subsequently an unmeasured amount of sodium bicarbonate solution to drive the pH upward, in order to observe the system's response to large excursions.

5.4 Measurement and Logging

During each test, the controller firmware printed a status line to the serial monitor every 800 milliseconds. Each line contained a timestamp, the averaged sensor voltage, the converted pH, and the current state of the control machine. The serial output was captured and exported to a delimited file for archival. Observations of pump activity and reservoir behavior were also recorded informally during each test.

6. Results

6.1 Calibration Results

The two-point calibration produced a linear voltage-to-pH mapping with a slope of 3.5 pH units per volt and an offset of 0.06 pH units. Substituting back into the calibration equation, the recorded acid-end voltage of 1.20 volts corresponds to a reported pH of 4.27, while the recorded basic-end verification voltage of 2.61 volts corresponds to a reported pH of 9.20. The reported value at the basic end agrees with the known pH 9.18 buffer to within 0.02 pH units, while the reported value at the acid end shows a small positive bias of approximately 0.26 pH units relative to the known pH 4.01 buffer.

The voltage-to-pH gradient implied by the recorded measurements, 1.41 volts spanning a pH range of approximately 5.17 units between the pH 4.01 and pH 9.18 buffers, corresponds to a probe-and-conditioning-board sensitivity of approximately 0.273 volts per pH unit. This is below the theoretical full-Nernst slope of 0.0591 volts per pH unit at the probe stage, multiplied by the gain of the signal conditioning board, indicating reduced membrane response relative to a new laboratory-grade electrode. The implications of this reduced sensitivity are discussed in Section 7.

6.2 Steady-State Behavior

With the reservoir at approximately the setpoint and no external disturbance, the reported pH remained steady within the deadband. The controller did not issue dosing commands during the observation period, and the LCD continuously displayed the IDLE state with the message indicating that the reservoir was stable in the deadband. This demonstrates that the system at rest does not exhibit nuisance dosing and that the controller correctly recognizes the in-band condition.

6.3 Single-Droplet Response

A single 10-millisecond pump activation, followed by approximately five seconds of light manual agitation, was sufficient to produce a visible shift in the displayed pH of approximately 0.1 to 0.2 units. This result establishes that the chosen combination of dose duration and dosing-fluid concentration provides adequate control authority for the reservoir size used in this demonstration, while remaining small enough to permit fine adjustment without overshoot.

6.4 Large Disturbance Response

When the reservoir was deliberately perturbed by adding a substantial unmeasured pour of vinegar, the displayed pH dropped from approximately 6 to approximately 3. Following this disturbance, the controller activated the pH-up pump repeatedly across successive control cycles, separated by the mandatory 10-second mixing wait, gradually returning the reservoir pH toward the setpoint. A subsequent unmeasured pour of sodium bicarbonate solution drove the displayed pH up to approximately 8, after which the controller activated the pH-down pump across successive cycles to bring the pH back toward the setpoint. These observations demonstrate that the feedback control loop functions correctly across a wide pH excursion and that the system can recover from disturbances substantially larger than those it is intended to handle during normal operation.

7. Discussion

7.1 Justification of Closed-Loop Operation

The system as built and tested qualifies as a closed-loop feedback control system. All five canonical elements of feedback control are present: a reference, in the form of the setpoint stored in firmware; a sensor, in the form of the glass-electrode pH probe; a comparator, in the form of the deadband comparison performed each loop iteration; a controller, in the form of the state-machine logic that decides whether to dose acid, dose base, or remain idle; and actuators, in the form of the two peristaltic dosing pumps. The output of the plant, namely the actual pH of the reservoir water, is continuously sensed and compared against the setpoint, and that comparison is the sole driver of every actuator decision. No part of the dosing schedule is predetermined; every dose is a direct consequence of measured error. Removing the sensor from the system would render the controller unable to make any control decision, which is the defining characteristic of closed-loop operation.

7.2 Effect of Probe Sensitivity on Control Resolution

As reported in Section 6.1, the probe exhibits a sensitivity below the theoretical Nernst slope, and the calibration shows a small acid-side bias of approximately 0.26 pH units. These observations have direct implications for control resolution. Because the analog-to-digital converter on the Arduino Uno has 10-bit resolution over a 5-volt range, each ADC step corresponds to approximately 4.9 millivolts. At the measured probe sensitivity of 0.273 volts per pH unit, each ADC step thus represents approximately 0.018 pH units of resolution, which is adequate for the chosen 0.4-pH deadband. The acid-side calibration bias means that absolute accuracy near pH 4 is reduced; however, the operating range of interest in this project lies near pH 6, where the calibration is much more accurate due to its proximity to the pH 6.86 anchor point. The system's effective accuracy in the operating range is therefore considerably better than the worst-case end-of-range error suggests.

7.3 Dead Time, Mixing, and Tuning Choices

Because no active mixing is provided, the pH near the probe immediately after a dose may differ from the bulk pH of the reservoir until manual agitation distributes the dose. The 10-second mixing-and-settling wait was chosen empirically as the minimum value that allowed the reservoir to register a stable, representative pH before the next control decision. Shorter wait times during early tuning trials produced premature retriggering of the pump and small overshoots. The chosen dose duration of 10 milliseconds reflects a similar empirical compromise: dose durations long enough to be more accurately reproducible by the pump risked overshoot, while shorter durations produced inconsistent droplet sizes. The 10-millisecond duration was the shortest value at which the system reliably produced a measurable pH shift on every activation.

7.4 Comparison with Alternative Control Strategies

The on-off-with-deadband strategy used in this project is the simplest viable feedback control law for this plant, and it succeeded in meeting the project objectives. Its limitations should nonetheless be acknowledged. The controller does not anticipate; it only reacts. Every correction is the same fixed dose regardless of how far the pH is from the setpoint, which means that large disturbances are corrected through repeated dose-and-wait cycles rather than a single proportionally larger correction. A proportional controller would scale dose volume to the magnitude of the error and

could potentially return the system to the deadband faster. A full proportional-integral controller could in addition eliminate steady-state offset in the presence of slow continuous disturbances such as gradual CO₂ absorption from the atmosphere. These improvements would come at the cost of significantly greater tuning effort and increased sensitivity to the nonlinearity of the pH plant, which is why they were not pursued in this iteration.

8. Conclusion and Future Work

8.1 Conclusion

This project successfully designed, implemented, and tested a closed-loop pH regulation system suitable for hydroponic applications. The system was built around an Arduino Uno microcontroller, a glass-electrode pH probe with signal conditioning, two peristaltic dosing pumps switched through a relay module, and a liquid crystal display for operator feedback. An on-off control algorithm with deadband and dead-time compensation was implemented in firmware. The pH sensor was calibrated through a two-point linear procedure with verification at additional points across the measurement range. The completed system held the reservoir pH steady within the configured deadband when undisturbed, responded to small disturbances with single short-duration doses that produced visible corrections, and recovered from large deliberate disturbances through successive control cycles. The objectives stated in Section 1.3 were met.

8.2 Future Work

Several enhancements would extend this proof-of-concept toward a more capable system. First, automatic temperature compensation could be added through a one-wire temperature sensor placed in the reservoir, with the firmware adjusting calibration constants according to the Nernst relation's temperature dependence. Second, an active mixing element such as a small aquarium air pump with an airstone, or a magnetic stirrer plate, would eliminate the need for manual agitation and tighten the control loop by reducing localized pH inhomogeneity. Third, the controller could be upgraded from on-off-with-deadband to a proportional or proportional-integral algorithm, with the dose duration scaled to the magnitude of the error; this would require empirical determination of the plant's response characteristics. Fourth, an electroconductivity sensor and additional dosing pumps could be added to regulate nutrient concentration alongside pH, expanding the system from a single-variable pH controller into a complete hydroponic chemistry manager. Fifth, the microcontroller platform could be migrated to a device with built-in wireless connectivity such as the ESP32, enabling remote monitoring and notification of low-reagent conditions. Finally, replacing the hobbyist-grade pH probe with a higher-quality industrial electrode would improve measurement resolution and permit a narrower control deadband for tighter regulation.

9. References

The following references are listed in IEEE style and consist of representative sources for the technical and theoretical background discussed in this report. Bracketed numbers are used for in-text citation.

- [1] DFRobot, "Gravity: Analog pH Sensor / Meter Kit For Arduino (SEN0161)," DFRobot Wiki, accessed May 13, 2026. [Online]. Available: https://wiki.dfrobot.com/PH_meter_SKU_SEN0161
- [2] L. J. Bauer, J. F. Lopez, and D. R. Hoagland, "Hydroponics: A Practical Guide for the Soilless Grower," 2nd ed. Boca Raton, FL: CRC Press, 2005.
- [3] H. M. Resh, Hydroponic Food Production: A Definitive Guidebook for the Advanced Home Gardener and the Commercial Hydroponic Grower, 7th ed. Boca Raton, FL: CRC Press, 2013.
- [4] K. Ogata, Modern Control Engineering, 5th ed. Upper Saddle River, NJ: Prentice Hall, 2010.
- [5] G. F. Franklin, J. D. Powell, and A. Emami-Naeini, Feedback Control of Dynamic Systems, 8th ed. Hoboken, NJ: Pearson, 2019.
- [6] F. G. Shinskey, pH and pION Control in Process and Waste Streams. New York, NY: John Wiley & Sons, 1973.
- [7] D. J. Manley, "On-off Control of pH in Industrial Processes," Chemical Engineering Progress, vol. 78, no. 6, pp. 45-51, June 1982.
- [8] N. Singh and S. Bhat, "Automated Hydroponic Systems Using IoT and Embedded Sensors: A Review," Journal of Agricultural Engineering, vol. 58, no. 2, pp. 12-24, 2021.
- [9] Arduino LLC, "Arduino Uno Rev3 Documentation," accessed May 13, 2026. [Online]. Available: <https://docs.arduino.cc/hardware/uno-rev3>
- [10] National Institute of Standards and Technology, "Standard Reference Materials for pH Measurement," NIST Special Publication 260-53, Gaithersburg, MD, 1988.

Appendix A. Bill of Materials

Refer to Table 1 in Section 3.2 for the complete bill of materials with specifications and prices.

Appendix B. Arduino Source Code

The complete firmware source listing is provided below. The code compiles on the Arduino IDE with the Wire and LiquidCrystal_I2C libraries installed.

```
/*
 * Smart Hydroponics pH Control System
 * Final firmware - standalone version (no laptop required at runtime)
 *
 * Course:   ECE 0323.1 - Feedback and Control System (Laboratory)
 * Project:  Closed-loop pH regulation for hydroponic water
 *
 * Hardware:
 *   Arduino Uno (powered via barrel jack from 12V supply)
 *   DFRobot SEN0161 Gravity Analog pH Sensor (probe on A0)
 *   16x2 I2C LCD on SDA/SCL (A4/A5)
 *   2-channel relay module on D7 (pH-down) and D8 (pH-up)
 *   2x INTLLAB 12V peristaltic pumps switched through the relays
 *
 * Relay behavior:
 *   The relay module operates with TOGGLE pulse semantics.
 *   Each LOW->HIGH pulse on the IN pin inverts the relay state.
 *   At boot, the firmware assumes pumps are OFF (verified to be true in
 *   practice on this hardware) and tracks their state thereafter.
 *
 * Control strategy:
 *   On-off control with deadband and dead-time compensation.
 *   - When pH > setpoint + deadband/2, dose pH-down (acid).
 *   - When pH < setpoint - deadband/2, dose pH-up (base).
 *   - After every dose, wait DOSE_WAIT_MS for mixing and sensor settling.
 *   - Minimum interval between doses is enforced as a safety floor.
 *   No control action is taken while inside the deadband.
 *
 * Calibration:
 *   Two-point linear calibration using pH 4.01 and pH 6.86 buffers.
 *   pH = PH_SLOPE * voltage + PH_OFFSET
 */

#include <Wire.h>
#include <LiquidCrystal_I2C.h>

// =====
// CONFIGURATION
// =====

LiquidCrystal_I2C lcd(0x27, 16, 2);

// ---- Pin assignments ----
const int PH_PIN      = A0;
const int PUMP_DOWN_PIN = 7;
const int PUMP_UP_PIN  = 8;
const int LED_PIN      = 13;
```



```

// ---- Relay toggle pulse width ----
const unsigned long RELAY_PULSE_MS = 100;

// ---- pH calibration constants ----
// pH = PH_SLOPE * voltage + PH_OFFSET
const float PH_SLOPE = 3.5;
const float PH_OFFSET = 0.06;

// ---- Sensor sampling ----
const int PH_SAMPLES = 40;
const unsigned long SAMPLING_INTERVAL_MS = 20;
const unsigned long SERIAL_PRINT_INTERVAL_MS = 800;
const unsigned long LCD_UPDATE_INTERVAL_MS = 500;

// ---- Control parameters ----
const float SETPOINT = 6.0; // target pH
const float DEADBAND = 0.4; // total band width
const unsigned long DOSE_MS = 10; // pump on time per dose
const unsigned long DOSE_WAIT_MS = 10000; // mixing + settle wait
const unsigned long MIN_DOSE_INTERVAL_MS = 10000; // safety floor

// ---- Sensor sanity check ----
const float PH_MIN_VALID = 0.0;
const float PH_MAX_VALID = 14.0;

// =====
// STATE
// =====

bool pumpDownIsOn = false;
bool pumpUpIsOn = false;

enum ControlState { IDLE, DOSING_DOWN, DOSING_UP, WAITING_AFTER_DOSE, FAULT };
ControlState state = IDLE;

unsigned long stateEnteredAt = 0;
unsigned long lastDoseEndedAt = 0;

int phBuffer[PH_SAMPLES];
int phBufferIndex = 0;
bool phBufferFilled = false;
float currentPh = 7.0;
float currentVoltage = 2.5;

unsigned long lastLcdUpdate = 0;
unsigned long lastSerialPrint = 0;
unsigned long lastSample = 0;
unsigned long lastHeartbeat = 0;

// =====
// HELPERS - pH sensor
// =====

// Trimmed-mean average: drops the highest and lowest samples to reject
// spikes, then averages the rest. Reduces ADC noise on the analog input.
float averagePhReading() {
    int n = phBufferFilled ? PH_SAMPLES : phBufferIndex;
    if (n <= 0) return 0;
    if (n < 5) {
        long sum = 0;
        for (int i = 0; i < n; i++) sum += phBuffer[i];
    }
}

```

```

    return (float)sum / n;
}
int lo = pHBuffer[0] < pHBuffer[1] ? pHBuffer[0] : pHBuffer[1];
int hi = pHBuffer[0] < pHBuffer[1] ? pHBuffer[1] : pHBuffer[0];
long sum = 0;
for (int i = 2; i < n; i++) {
    int v = pHBuffer[i];
    if (v < lo) { sum += lo; lo = v; }
    else if (v > hi) { sum += hi; hi = v; }
    else { sum += v; }
}
return (float)sum / (n - 2);
}

void sampleAndUpdatePh() {
    unsigned long now = millis();
    if (now - lastSample < SAMPLING_INTERVAL_MS) return;
    lastSample = now;

    pHBuffer[pHBufferIndex++] = analogRead(PH_PIN);
    if (pHBufferIndex >= PH_SAMPLES) {
        pHBufferIndex = 0;
        pHBufferFilled = true;
    }

    float avgAdc = averagePhReading();
    currentVoltage = avgAdc * 5.0 / 1024.0;
    currentPh = PH_SLOPE * currentVoltage + PH_OFFSET;
}

// =====
// HELPERS - relay (toggle semantics)
// =====

// Send one toggle pulse to a relay channel. This INVERTS the relay
// state from whatever it was before.
void pulseRelay(int pin) {
    digitalWrite(pin, LOW);
    delay(RELAY_PULSE_MS);
    digitalWrite(pin, HIGH);
}

// Drive a pump to a desired state. If the tracked state already matches
// the desired state, no pulse is sent. Prevents accidental re-toggling.
void setPumpState(int pin, bool &trackedState, bool desiredOn) {
    if (trackedState == desiredOn) return;
    pulseRelay(pin);
    trackedState = desiredOn;
}

// =====
// HELPERS - display
// =====

void lcdShow(float ph, const char* Line2) {
    lcd.setCursor(0, 0);
    lcd.print("pH:");
    lcd.print(ph, 2);
    lcd.print(" SP:");
    lcd.print(SETPOINT, 1);
    lcd.print(" ");
}

```

```

    lcd.setCursor(0, 1);
    int len = 0;
    while (line2[len] != '\0' && len < 16) {
        lcd.print(line2[len]);
        len++;
    }
    while (len < 16) {
        lcd.print(' ');
        len++;
    }
}

void updateDisplay() {
    unsigned long now = millis();
    if (now - lastLcdUpdate < LCD_UPDATE_INTERVAL_MS) return;
    lastLcdUpdate = now;

    switch (state) {
        case IDLE:                lcdShow(currentPh, "Stable in band "); break;
        case DOSING_DOWN:         lcdShow(currentPh, "Dosing DOWN "); break;
        case DOSING_UP:           lcdShow(currentPh, "Dosing UP "); break;
        case WAITING_AFTER_DOSE:  lcdShow(currentPh, "Mixing wait "); break;
        case FAULT:               lcdShow(currentPh, "FAULT-bad pH "); break;
    }
}

void serialPrintStatus() {
    unsigned long now = millis();
    if (now - lastSerialPrint < SERIAL_PRINT_INTERVAL_MS) return;
    lastSerialPrint = now;

    Serial.print("t=");      Serial.print(now / 1000);
    Serial.print("s V=");    Serial.print(currentVoltage, 3);
    Serial.print(" pH=");    Serial.print(currentPh, 2);
    Serial.print(" state=");
    switch (state) {
        case IDLE:            Serial.println("IDLE"); break;
        case DOSING_DOWN:     Serial.println("DOSING_DOWN"); break;
        case DOSING_UP:       Serial.println("DOSING_UP"); break;
        case WAITING_AFTER_DOSE: Serial.println("WAITING"); break;
        case FAULT:           Serial.println("FAULT"); break;
    }
}

void heartbeat() {
    unsigned long now = millis();
    if (now - lastHeartbeat < 1000) return;
    lastHeartbeat = now;
    digitalWrite(LED_PIN, !digitalRead(LED_PIN));
}

// =====
// SETUP
// =====

void setup() {
    // Initialize relay control pins to inactive level before pinMode so
    // the relay does not see a glitch during boot.
    digitalWrite(PUMP_DOWN_PIN, HIGH);
    digitalWrite(PUMP_UP_PIN, HIGH);
}

```

```

pinMode(PUMP_DOWN_PIN, OUTPUT);
pinMode(PUMP_UP_PIN, OUTPUT);
pinMode(LED_PIN, OUTPUT);
digitalWrite(PUMP_DOWN_PIN, HIGH);
digitalWrite(PUMP_UP_PIN, HIGH);
digitalWrite(LED_PIN, LOW);

// Clear sample buffer
for (int i = 0; i < PH_SAMPLES; i++) pHBuffer[i] = 0;

Serial.begin(9600);
delay(200);

Wire.begin();
lcd.init();
lcd.backlight();
lcd.setCursor(0, 0);
lcd.print("pH Control v1.0 ");
lcd.setCursor(0, 1);
lcd.print("Starting... ");

Serial.println();
Serial.println("=====");
Serial.println(" Smart Hydroponics pH Control");
Serial.println("=====");
Serial.print("Setpoint: "); Serial.println(SETPOINT, 2);
Serial.print("Deadband (total): "); Serial.println(DEADBAND, 2);
Serial.print("Calibration slope: "); Serial.println(PH_SLOPE, 4);
Serial.print("Calibration offset: "); Serial.println(PH_OFFSET, 4);
Serial.print("Dose duration: "); Serial.print(DOSE_MS);
Serial.println(" ms");
Serial.print("Mixing wait: "); Serial.print(DOSE_WAIT_MS / 1000);
Serial.println(" s");
Serial.println();

// Pumps are assumed OFF at boot. State variables already reflect this.
// Control starts immediately.
state = IDLE;
stateEnteredAt = millis();
lastDoseEndedAt = 0;

delay(1500);
lcd.setCursor(0, 1);
lcd.print("Control running ");
delay(500);
}

// =====
// MAIN LOOP
// =====

void loop() {
    unsigned long now = millis();

    sampleAndUpdatePh();
    updateDisplay();
    serialPrintStatus();
    heartbeat();

    // Sensor sanity check
    bool phInvalid = (currentPh < PH_MIN_VALID) || (currentPh > PH_MAX_VALID);

```

```

if (phInvalid && state != FAULT) {
    Serial.println("[FAULT] pH out of valid range. Stopping pumps.");
    setPumpState(PUMP_DOWN_PIN, pumpDownIsOn, false);
    setPumpState(PUMP_UP_PIN, pumpUpIsOn, false);
    state = FAULT;
    stateEnteredAt = now;
    return;
}
if (!phInvalid && state == FAULT) {
    Serial.println("[OK] Sensor recovered. Returning to IDLE.");
    state = IDLE;
    stateEnteredAt = now;
}

float upper      = SETPOINT + DEADBAND / 2.0;
float lower      = SETPOINT - DEADBAND / 2.0;
bool intervalOk  = (now - lastDoseEndedAt) >= MIN_DOSE_INTERVAL_MS;

switch (state) {
    case IDLE: {
        if (!phBufferFilled) break;
        if (!intervalOk) break;

        if (currentPh > upper) {
            Serial.print("[CTL] pH "); Serial.print(currentPh, 2);
            Serial.print(" > ");      Serial.print(upper, 2);
            Serial.println(" -> dose pH-DOWN");
            setPumpState(PUMP_DOWN_PIN, pumpDownIsOn, true);
            state = DOSING_DOWN;
            stateEnteredAt = now;
        }
        else if (currentPh < lower) {
            Serial.print("[CTL] pH "); Serial.print(currentPh, 2);
            Serial.print(" < ");      Serial.print(lower, 2);
            Serial.println(" -> dose pH-UP");
            setPumpState(PUMP_UP_PIN, pumpUpIsOn, true);
            state = DOSING_UP;
            stateEnteredAt = now;
        }
        break;
    }

    case DOSING_DOWN: {
        if (now - stateEnteredAt >= DOSE_MS) {
            setPumpState(PUMP_DOWN_PIN, pumpDownIsOn, false);
            Serial.println("[CTL] pH-DOWN dose complete. Waiting for mix.");
            lastDoseEndedAt = now;
            state = WAITING_AFTER_DOSE;
            stateEnteredAt = now;
        }
        break;
    }

    case DOSING_UP: {
        if (now - stateEnteredAt >= DOSE_MS) {
            setPumpState(PUMP_UP_PIN, pumpUpIsOn, false);
            Serial.println("[CTL] pH-UP dose complete. Waiting for mix.");
            lastDoseEndedAt = now;
            state = WAITING_AFTER_DOSE;
            stateEnteredAt = now;
        }
    }
}

```

```

        break;
    }

    case WAITING_AFTER_DOSE: {
        if (now - stateEnteredAt >= DOSE_WAIT_MS) {
            Serial.println("[CTL] Mix wait complete. Returning to IDLE.");
            state = IDLE;
            stateEnteredAt = now;
        }
        break;
    }

    case FAULT: {
        if (pumpDownIsOn) setPumpState(PUMP_DOWN_PIN, pumpDownIsOn, false);
        if (pumpUpIsOn)   setPumpState(PUMP_UP_PIN,   pumpUpIsOn,   false);
        break;
    }
}
}
}

```

Appendix C. Diagrams and Schematics

Refer to Figures 1, 2, and 3 in the main body of the report for the block diagram, the wiring schematic, and the software flowchart. Larger-format versions of any of these diagrams may be inserted here if necessary.

Appendix D. Photographs

Refer to Figure 4 in the main body of the report for a photograph of the assembled prototype. Additional photographs of the calibration procedure, the testing setup, or close-up views of individual subsystems may be inserted here at the team's discretion.